

MCY 612 Cloud Security Assignment 4

Abstract

This essay explores various AWS storage solutions such as EC2 instance store, EBS, S3, and EFS and examines the key techniques and best practices for ensuring data protection across these services.

Introduction

In today's digital world, data has become one of the most valuable assets for individuals and organizations alike. From customer information and financial records to internal communications and operational systems, data powers nearly every part of modern business. As our reliance on cloud services, such as Amazon Web Services (AWS), continues to grow, so does the importance of keeping this data secure—whether it's being stored, transmitted, or processed. With cyber threats like ransomware, data breaches, and unauthorized access becoming more sophisticated and frequent, securing data has never been more critical. The risks associated with compromised data can lead to significant financial losses, regulatory penalties, and damage to a company's reputation.

This paper explores key concepts in cloud data security, focusing on how AWS helps protect data at rest and in transit using tools like AWS Key Management Service (KMS), encryption, and secure access controls. It also covers the different types of AWS storage options, including EBS, S3, and EFS, highlighting their unique security features. Moreover, it explains the role of encryption—both symmetric and asymmetric—in securing sensitive data, and how backup strategies are essential in defending against threats like ransomware attacks. By understanding and applying these security best practices, organizations can better protect their digital assets and ensure data remains safe, accessible, and resilient against potential cyber threats.

Protecting Data at Rest

Data at Rest includes any data that is being held on non-volatile storage, data that isn't actively processing or traveling over networks is referred to as data at rest. This includes data stored on IoT devices as well as databases, archives, and files in block or object storage. It may appear that this data is safe because it is in storage, but it still needs protection. This is due to the fact that it may be vulnerable to illegal access, particularly if it includes private correspondence, financial information, or client records. To lower the chance of unauthorized access or theft, data must be protected when it is at rest.

An example of Data at Rest is a business keeping client credit card numbers in its database is a realistic example. A major data breach, financial fraud, and legal issues could result from poorly protected data that is accessed by unauthorized individuals. Through services like AWS Key Management Service (KMS), AWS provides encryption to help protect data when at rest. Users can use this to encrypt their data, making it inaccessible without the right key. Additionally, AWS records each KMS key usage in AWS CloudTrail, allowing enterprises to audit and track access to make sure that only allowed activities are occurring.

Protecting Data in Transit

Data in Transit includes any data that is actively traveling from one system to another is referred to as data in transit. This includes data being transmitted between your systems and users or data being exchanged between services inside your AWS environment. This covers everything, from a file being sent from a server to a client to an API call. This data is susceptible to being intercepted or altered because it is

in motion. To ensure that data remains private and hasn't been changed during transmission, that is why securing it is so crucial.

An example of Data in Transit is when a customer enters their login credentials on a website, illustrates why data in transit needs to be protected. A hacker might intercept that data and take the credentials if it isn't secured. With solutions like AWS PrivateLink, which establishes safe, private connections between your VPCs or between on-premises systems and AWS services, AWS protects data while it's in transit. Your data remains on the secure Amazon network when you use PrivateLink, never going over the open internet. In addition to improving security, it also helps in meeting strict compliance standards such as EU/US Privacy Shield and HIPAA. Additionally, it eliminates the need to set up intricate firewall or routing rules, which makes it safer and simpler to administer.

Cryptography and AWS Key Management Service

The cryptographic technique called symmetric encryption uses the same key to encrypt and decrypt data. This implies that the exact same secret key must be available to both the sender and the recipient. The Advanced Encryption Standard is a known example of symmetric encryption (AES). Let's say Jane wishes to communicate with John in a private way. She encrypts her message using AES and a shared secret key. John uses the same key to decrypt the ciphertext and read the original message after receiving it. One of the main problems with symmetric encryption is safely distributing the key in the first place because it needs to be kept secret. In contrast, asymmetric encryption makes use of a public key and a private key. While the private key is kept private and is used for decoding data, the public key is widely available and is used to encrypt data. RSA encryption is one example. Jane can encrypt the message using John's public key if she wants to send John a secure communication. The message can only be read by John since only John's private key can decode it. This technique, which is slower and typically used to encrypt small bits of data or securely exchange symmetric keys, resolves the key distribution issue of symmetric encryption.

An EC2 key pair, consisting of a public and a private key, is created when you start a Linux EC2 instance on AWS. The private key (a .pem file) is downloaded, and the public key is saved on the EC2 instance upon launch. For instance, your local computer verifies with the EC2 instance using the private key when you login to it over SSH. The EC2 server verifies that the request matches its public key before granting access. The EC2 instance takes on the role of John in this real-world example of asymmetric encryption, ensuring that only you, who has the correct private key, can access it.

The managed service AWS Key Management Service (KMS) assists you in generating and managing the cryptographic keys that protect your information. It enables centralized encryption control and connects with numerous AWS services. Because KMS employs envelope encryption, your data is first encrypted using a data key, which is subsequently encrypted using a master key that is controlled by KMS. For instance, a data key is used to encrypt the file when you upload it to S3 and activate server-side encryption with AWS KMS. Your KMS key is then used to encrypt the data key. This method reduces the exposure of your primary encryption keys, which enhances security and performance.

Protecting Storage

Amazon EC2 offers a number of storage choices, including Instance Store, EBS, S3, and EFS, each of which is suited to a particular use case. Temporary block storage that is physically connected to an EC2 instance is called an instance store. When an instance is stopped, hibernated, or terminated, the data stored on the instance store is lost. Data that doesn't need to exist past an instance's lifecycle can benefit from it. On the other hand, EC2 instances can be connected to EBS (Elastic Block Store), which offers persistent block-level storage that is kept even if the instance is stopped or terminated. EBS volumes can be backed up using snapshots and encrypted for further protection. For a variety of data formats, S3 (Simple Storage Service) provides incredibly strong object storage. With several storage classes to suit different demands, it is perfect for storing massive amounts of unstructured data. Multiple instances can access the same file system at once thanks to EFS's (Elastic File System) scalable file storage for EC2 instances. EFS is especially helpful for programs that need a shared file system.

AWS uses encryption and safe data storage procedures to protect EBS. By default, AWS Key Management Service (KMS) keys are used to encrypt volumes and snapshots. EBS protects data from unwanted access by encrypting it while it's in transit and at rest. The XTS-AES-256 cipher is used to encrypt instance store volumes, including those on NVMe devices, and encryption keys are safely stored inside the instance's hardware module. The keys are destroyed when instances are stopped or terminated, and any sensitive data is cleared out before reusing to avoid unwanted access.

AWS provides a very strong architecture for S3 that includes security features like object lock and data versioning to stop accidental or intentional deletions. With data redundantly stored across at least three Availability Zones, S3 provides 99.999999999% durability. Versioning enables users to recover from program or human errors by preserving, retrieving, and restoring earlier versions of items. AWS advises encrypting sensitive data both in transit and at rest using SSL/TLS or AWS KMS and implementing strict IAM controls to manage access to S3 resources. S3 allows replication and versioning for improved security and cross-region backup for additional data protection.

EFS is protected by AWS using both encryption in transit and encryption at rest. Encryption at rest can be enabled when establishing an EFS file system, guaranteeing the security of the data contained inside. Data traveling between the file system and EC2 instances is protected by encryption in transit, which is accomplished by turning on TLS when mounting the file system. AWS suggests sticking to best practices, which include implementing multi-factor authentication (MFA) for accounts, using IAM for access control. In order to guarantee that encryption keys are handled and managed securely, Amazon EFS also makes use of the AWS KMS for key management.

Protecting Data w/ Backup

A data backup is a copy of your data that is kept separate and can be restored in the event that the original is compromised, lost, or corrupted. Backups are crucial for defending crucial data and systems against unplanned malfunctions, accidents, and malicious attacks. With AWS Backup, which supports a variety of services like EBS, S3, RDS, DynamoDB, and EFS, backups can be made automatically on a schedule or manually (on-demand). Configuring which resources to backup, establishing retention guidelines, and even scheduling backups for when new resources are produced are all possible with the AWS Backup console. AWS protects data by promoting encryption, multi-factor authentication (MFA), secure communication via SSL/TLS, and strict access controls with IAM.

A ransomware attack is a type of cyber threat where attackers use malicious software to gain unauthorized access to a system, encrypt important files, and block access to them until a ransom is paid. Essentially, the victim is locked out of their own data and systems, and the attacker demands payment—usually in cryptocurrency—in exchange for a decryption key. However, even if the ransom is paid, there's no guarantee that the data will be fully restored without corruption. These attacks can cause major disruptions, financial losses, and reputational damage. Some high-profile examples include Colonial Pipeline, CNA Financial, and Kia Motors, all of which suffered significant operational setbacks.

Having reliable and secure backups is one of the most effective ways to mitigate the impact of ransomware attacks. A well-structured backup strategy ensures that even if attackers encrypt or destroy primary data, organizations can restore clean versions from backup without paying a ransom. AWS offers powerful features like the Multi-Account Backup Method, where backups are stored in a separate, isolated AWS account. AWS Backup Vault Lock adds another layer of protection by making backups tamper-proof through a write-once, read-many (WORM) setting. Additionally, features like cold storage transition, retention policies, and strict access controls ensure that backup data remains protected, cost-efficient, and available for recovery even in the event of a ransomware breach.

Conclusion

In conclusion, protecting data in the cloud requires a strong understanding of where data exists and how it moves. Data at rest—such as information stored in databases, IoT devices, or AWS storage services like EBS, S3, and EFS—must be secured from unauthorized access. Likewise, data in transit—moving between systems or users—needs protection from interception or tampering. AWS offers

a comprehensive suite of tools to support this, including encryption through AWS Key Management Service (KMS), secure transmission via SSL/TLS, and access control through IAM and multi-factor authentication (MFA). These measures ensure that both storage and communication are tightly protected.

Techniques like symmetric and asymmetric encryption help maintain confidentiality, whether you're storing sensitive files or connecting securely to an EC2 instance. Implementing strong backup strategies is another crucial defense, particularly against ransomware attacks. Services like AWS Backup allow automated and on-demand backups across multiple AWS resources, with features like cross-account storage, vault locks, and retention policies to ensure data remains recoverable and protected. By combining encryption, access control, secure storage, and smart backup practices, organizations can build a resilient data security framework that guards against both internal risks and external threats in today's complex cloud environments.

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